UC Berkeley

Department of Electrical Engineering and Computer Sciences

EECS 127 - EECS 227AT. Optimization Models in Engineering

Lecture Note 6

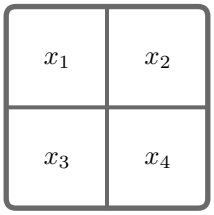
Fall 2025

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Recall the MRI example:



- There are 4 unknowns.
- If the number of beams is less than 4, we can't uniquely find the unknowns. It makes sense to use as many beams as possible to increase the data.
- This is the *overdetermined* case.
- The measurements are often noisy that makes equations inconsistent. As a result, $\mathbf{A}\boldsymbol{x} = \boldsymbol{y}$ won't have a solution.

Example 1. Let's say we have $m = 3$ measurements and $n = 2$ unknowns:

$$\begin{cases} x_1 + x_2 = 2, & \text{(1st measurement)} \\ 2x_1 + 3x_2 = 5, & \text{(2nd measurement)} \\ 4x_1 - x_2 = 3. & \text{(3rd measurement)} \end{cases}$$

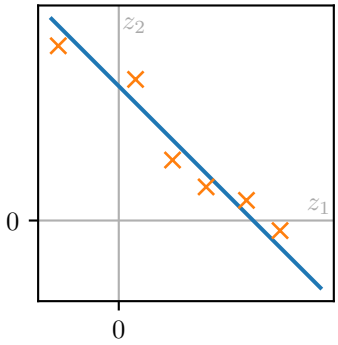
In this case there is a solution $\boldsymbol{x} = [1, 1]^{\mathrm{T}}$. Now, if 2 in the first measurement is changed to $2 + \varepsilon$, where ε is the noise, then there's no solution. ■

In the case where there is no solution, we solve the least-squares (LS) problem:

$$\min_{\boldsymbol{x}} \quad \|\mathbf{A}\boldsymbol{x} - \boldsymbol{y}\|.$$

If the optimal objective is zero, we have $\|\mathbf{A}\boldsymbol{x}^* - \boldsymbol{y}\| = 0 \Rightarrow \mathbf{A}\boldsymbol{x}^* = \boldsymbol{y}$ (solution exists). Otherwise, $\mathbf{A}\boldsymbol{x}^* - \boldsymbol{y} \neq \mathbf{0}$, and $\|\mathbf{A}\boldsymbol{x} - \boldsymbol{y}\|$ measures how much we violate the equation $\mathbf{A}\boldsymbol{x} = \boldsymbol{y}$.

Example 2 (Linear regression)**.** Consider a set of points $\boldsymbol{z}^{(i)} = [z_1^{(i)}, z_2^{(i)}]^{\mathrm{T}} \in \mathbb{R}^2$. We want to find a line that is close to all (or most) points.



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- A line can be written as: $z_2 = x_1 z_1 + x_2$. So, there are two unknown coefficients x_1 and x_2.
- Consider $z_2^{(i)} - x_1 z_1^{(i)} - x_2$ for $i = 1, \ldots, m$. Note that $z_2^{(i)} - x_1 z_1^{(i)} - x_2$ may not be zero. So, we minimize the sum of square of these residues:

$$\begin{aligned}\min_{\boldsymbol{x}\in\mathbb{R}^2} \sum_{i=1}^{m} \left(x_1 z_1^{(i)} + x_2 - z_2^{(i)}\right)^2 &= \min_{\boldsymbol{x}\in\mathbb{R}^2} \sum_{i=1}^{m} \left(\begin{bmatrix} z_1^{(i)} & 1 \end{bmatrix} \boldsymbol{x} - z_2^{(i)}\right)^2 \\ &= \min_{\boldsymbol{x}\in\mathbb{R}^2} \|\mathbf{A}\boldsymbol{x} - \boldsymbol{y}\|_2^2,\end{aligned}$$

  where

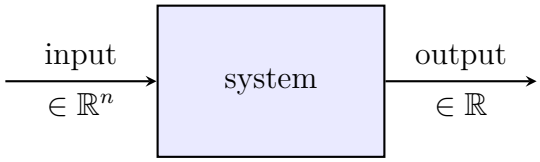
$$\mathbf{A} = \begin{bmatrix} z_1^{(1)} & 1 \\ \vdots & \vdots \\ z_1^{(m)} & 1 \end{bmatrix}, \quad \boldsymbol{y} = \begin{bmatrix} z_2^{(1)} \\ \vdots \\ z_2^{(m)} \end{bmatrix}.$$

- Note that $\min\|\mathbf{A}\boldsymbol{x} - \boldsymbol{y}\|$ is the same as $\min\|\mathbf{A}\boldsymbol{x} - \boldsymbol{y}\|^2$ because

$$\underbrace{\|\mathbf{A}\boldsymbol{x}^* - \boldsymbol{y}\|}_{\geq 0} \leq \underbrace{\|\mathbf{A}\boldsymbol{x} - \boldsymbol{y}\|}_{\geq 0}, \forall \boldsymbol{x} \quad \Longleftrightarrow \quad \underbrace{\|\mathbf{A}\boldsymbol{x}^* - \boldsymbol{y}\|}_{\geq 0}^2 \leq \underbrace{\|\mathbf{A}\boldsymbol{x} - \boldsymbol{y}\|}_{\geq 0}^2, \forall \boldsymbol{x}.$$

■

General linear regression: learn a linear system where the output is a linear function of the input (we learn the coefficients of the linear function).



- Let the input be $\boldsymbol{a} \in \mathbb{R}^n$, then the output is $\boldsymbol{a}^{\mathrm{T}}\boldsymbol{x}$, where $\boldsymbol{x}$ is a vector of coefficients.
- We apply inputs $\boldsymbol{a}^{(1)}, \ldots, \boldsymbol{a}^{(m)} \in \mathbb{R}^n$ and measure *noisy* output $y_1, \ldots, y_m \in \mathbb{R}$.
- The least-squares problem is

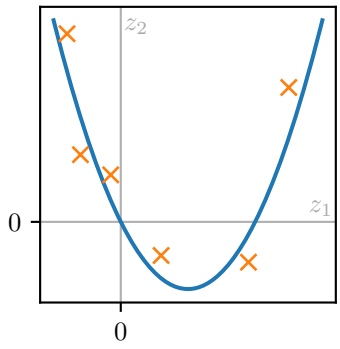
$$\min_{\boldsymbol{x}\in\mathbb{R}^n} \sum_{i=1}^{m} \left((\boldsymbol{a}^{(i)})^{\mathrm{T}}\boldsymbol{x} - y_i\right)^2 = \min_{\boldsymbol{x}\in\mathbb{R}^n} \|\mathbf{A}\boldsymbol{x} - \boldsymbol{y}\|^2,$$

  where

$$\mathbf{A} = \begin{bmatrix} (\boldsymbol{a}^{(1)})^{\mathrm{T}} \\ \vdots \\ (\boldsymbol{a}^{(m)})^{\mathrm{T}} \end{bmatrix} \in \mathbb{R}^{m\times n}, \quad \boldsymbol{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix} \in \mathbb{R}^m.$$

Example 3 (Non-linear regression)**.** Find a quadratic function to fit a set of points.

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- Points: $(z_1^{(i)}, z_2^{(i)})$, $i = 1, \ldots, m$.
- Model: $z_2 = x_1 z_1^2 + x_2 z_1 + x_3$, where x_1, x_2, x_3 are the unknown coefficients.
- The objective is

$$\min_{\boldsymbol{x}\in\mathbb{R}^3} \sum_{i=1}^{m} \left(x_1 (z_1^{(i)})^2 + x_2 z_1^{(i)} + x_3 - z_2^{(i)} \right)^2 = \min_{\boldsymbol{x}\in\mathbb{R}^3} \|\mathbf{A}\boldsymbol{x} - \boldsymbol{y}\|^2,$$

where

$$\mathbf{A} = \begin{bmatrix} (z_1^{(i)})^2 & z_1^{(1)} & 1 \\ \vdots & \vdots & \vdots \\ (z_1^{(m)})^2 & z_1^{(m)} & 1 \end{bmatrix}, \quad \boldsymbol{y} = \begin{bmatrix} z_2^{(1)} \\ \vdots \\ z_2^{(m)} \end{bmatrix}.$$

■

- Why minimize $\|\mathbf{A}\boldsymbol{x} - \boldsymbol{y}\|_2$ and not another function of error such as $\|\mathbf{A}\boldsymbol{x} - \boldsymbol{y}\|_1$ or $\|\mathbf{A}\boldsymbol{x} - \boldsymbol{y}\|_\infty$?
- Assume measured $\boldsymbol{y}$ = correct $\tilde{\boldsymbol{y}}$ + sensor noise $\bar{\boldsymbol{y}}$.
- For example, in the MRI problem, we may have $y_1 = 2 + \varepsilon$. Then, $\tilde{y}_1 = 2$, $\bar{y}_1 = \varepsilon$.
- If $\bar{\boldsymbol{y}}$ is Gaussian and $\bar{y}_1, \ldots, \bar{y}_m$ are i.i.d. (independent, identically distributed), it turns out that the least-squares solution gives the best estimation of the correct unknown model given data.
- In statistics, it is said that the lease-square solution is the *maximum likelihood estimator* (MLE) of the ground-truth model $\boldsymbol{x}$.

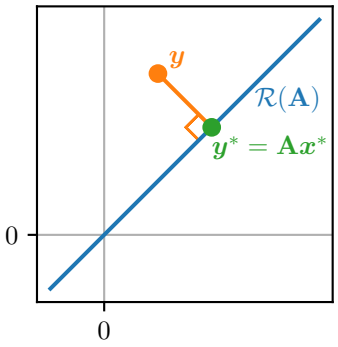
How to solve

$$\min_{\boldsymbol{x}\in\mathbb{R}^n} \|\mathbf{A}\boldsymbol{x} - \boldsymbol{y}\|. \qquad (\mathbf{A} \in \mathbb{R}^{m\times n})$$

- The set $\mathcal{R}(\mathbf{A}) = \{\mathbf{A}\boldsymbol{x} \mid \boldsymbol{x} \in \mathbb{R}^n\}$ is a subspace.
- The solution to $\min\|\mathbf{A}\boldsymbol{x} - \boldsymbol{y}\|$ corresponds to the projection of $\boldsymbol{y}$ on $\mathcal{R}(\mathbf{A})$.

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- Let the projected point be called $\boldsymbol{y}^*$.
- $\boldsymbol{y}^*$ exists and is unique.
- Theorem on projection: $(\boldsymbol{y} - \boldsymbol{y}^*) \perp \mathcal{R}(\mathbf{A})$.



- Also, since $\mathcal{N}(\mathbf{A}^{\mathrm{T}}) \perp \mathcal{R}(\mathbf{A})$ and $\mathcal{N}(\mathbf{A}^{\mathrm{T}}) \oplus \mathcal{R}(\mathbf{A}) = \mathbb{R}^m$, we have

$$\boldsymbol{y} - \boldsymbol{y}^* \in \mathcal{N}(\mathbf{A}^{\mathrm{T}}) \quad \Longrightarrow \quad \mathbf{A}^{\mathrm{T}}(\boldsymbol{y} - \boldsymbol{y}^*) = \mathbf{0}.$$

- Since $\boldsymbol{y}^* \in \mathcal{R}(\mathbf{A})$, there is at least one $\boldsymbol{x}^*$ such that $\mathbf{A}\boldsymbol{x}^* = \boldsymbol{y}^*$. Thus,

$$\mathbf{0} = \mathbf{A}^{\mathrm{T}}(\boldsymbol{y} - \boldsymbol{y}^*) = \mathbf{A}^{\mathrm{T}}(\boldsymbol{y} - \mathbf{A}\boldsymbol{x}^*) \quad \Longrightarrow \quad \mathbf{A}^{\mathrm{T}}\boldsymbol{y} = \mathbf{A}^{\mathrm{T}}\mathbf{A}\boldsymbol{x}^*.$$

- Note that while $\boldsymbol{y}^*$ is unique, there may be infinitely many $\boldsymbol{x}^*$ associated with $\boldsymbol{y}^*$.

Theorem 1 (Solution to least-squares)**.** *The least-squares problem always has a solution, and the set of solutions is*

$$\{\boldsymbol{x}^* \mid \mathbf{A}^{\mathrm{T}}\mathbf{A}\boldsymbol{x}^* = \mathbf{A}^{\mathrm{T}}\boldsymbol{y}\}.$$

If $\mathbf{A}$ *has full column rank (which requires* $m \geq n$*), then* $\mathbf{A}^{\mathrm{T}}\mathbf{A}$ *is invertible, and*

$$\boldsymbol{x}^* = \underbrace{\boxed{(\mathbf{A}^{\mathrm{T}}\mathbf{A})^{-1}\mathbf{A}^{\mathrm{T}}}}_{\text{pseudoinverse}}\boldsymbol{y} = \mathbf{A}^{+}\boldsymbol{y}$$

is the unique solution.

(Note: if $\mathbf{A}$ has full column rank, $\mathbf{A}^{+} = (\mathbf{A}^{\mathrm{T}}\mathbf{A})^{-1}\mathbf{A}^{\mathrm{T}}$; if $\mathbf{A}$ has full row rank, $\mathbf{A}^{+} = \mathbf{A}^{\mathrm{T}}(\mathbf{A}\mathbf{A}^{\mathrm{T}})^{-1}$.)

Another way to obtain this formula:

$$\min \ f(\boldsymbol{x}) \quad \longrightarrow \quad \nabla f(\boldsymbol{x}^*) = \mathbf{0}.$$

- The issue is that $\|\mathbf{A}\boldsymbol{x} - \boldsymbol{y}\|$ is a norm function and $\|\cdot\|$ is not differentiable at the origin.

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- However, since $\min\|\mathbf{A}\boldsymbol{x}-\boldsymbol{y}\|$ is equivalent to $\min\|\mathbf{A}\boldsymbol{x}-\boldsymbol{y}\|^2$, we can minimize the latter instead:

$$\begin{aligned}\|\mathbf{A}\boldsymbol{x}-\boldsymbol{y}\|^2 &= \langle \mathbf{A}\boldsymbol{x}-\boldsymbol{y}, \mathbf{A}\boldsymbol{x}-\boldsymbol{y}\rangle \\ &= (\mathbf{A}\boldsymbol{x}-\boldsymbol{y})^{\mathrm{T}}(\mathbf{A}\boldsymbol{x}-\boldsymbol{y}) \\ &= (\boldsymbol{x}^{\mathrm{T}}\mathbf{A}^{\mathrm{T}}-\boldsymbol{y}^{\mathrm{T}})(\mathbf{A}\boldsymbol{x}-\boldsymbol{y}) \\ &= \boldsymbol{x}^{\mathrm{T}}\mathbf{P}\boldsymbol{x}+\boldsymbol{q}^{\mathrm{T}}\boldsymbol{x}+r,\end{aligned}$$

  where $\mathbf{P}=\mathbf{A}^{\mathrm{T}}\mathbf{A}\in\mathbb{R}^{n\times n}$, $\boldsymbol{q}=-2\mathbf{A}^{\mathrm{T}}\boldsymbol{y}\in\mathbb{R}^n$, $r=\boldsymbol{y}^{\mathrm{T}}\boldsymbol{y}\in\mathbb{R}$.

A function of the form $\boldsymbol{x}^{\mathrm{T}}\mathbf{P}\boldsymbol{x}+\boldsymbol{q}^{\mathrm{T}}\boldsymbol{x}+r$ is called a *quadratic function*.

Example 4. Example for $n=1$:

$$f(x)=2x^2+5x-1.$$

A quadratic function with one variable can be written as

$$px^2+qx+r.$$

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Example 5. Example for $n=2$:

$$\begin{aligned}f(\boldsymbol{x}) &= x_1^2+5x_2^2-6x_1x_2+7x_1+2x_2-5 \\ &= \begin{bmatrix}x_1 & x_2\end{bmatrix}\underbrace{\begin{bmatrix}1 & -3\\ -3 & 5\end{bmatrix}}_{\mathbf{P}}\begin{bmatrix}x_1\\ x_2\end{bmatrix}+\underbrace{\begin{bmatrix}7 & 2\end{bmatrix}}_{\boldsymbol{q}^{\mathrm{T}}}\boldsymbol{x}+\underbrace{(-5)}_{r}.\end{aligned}$$

■

Notice that in Example 5, $\mathbf{P}$ can also be chosen as $\begin{bmatrix}1 & -6\\ 0 & 5\end{bmatrix}$ or $\begin{bmatrix}1 & -4\\ -2 & 5\end{bmatrix}$. But, in the quadratic form, we always choose $\mathbf{P}$ to be *symmetric*, i.e., $\mathbf{P}=\mathbf{P}^{\mathrm{T}}$ (reason to be discussed later).
Let the (i,j)-th entry of $\mathbf{P}$ be denoted as P_{ij}. We have

$$\begin{aligned}f(\boldsymbol{x}) &= \boldsymbol{x}^{\mathrm{T}}\mathbf{P}\boldsymbol{x}+\boldsymbol{q}^{\mathrm{T}}\boldsymbol{x}+r \\ &= \sum_{i=1}^{n}\sum_{j=1}^{n}P_{ij}x_ix_j+\sum_{i=1}^{n}q_ix_i+r \\ &= \sum_{i=1}^{n}P_{ii}x_i^2+\sum_{i<j}(P_{ij}+P_{ji})x_ix_j+\sum_{i=1}^{n}q_ix_i+r \\ &= \sum_{i=1}^{n}P_{ii}x_i^2+2\sum_{i<j}P_{ij}x_ix_j+\sum_{i=1}^{n}q_ix_i+r,\end{aligned}$$

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where we used $P_{ij} = P_{ji}$ since $\mathbf{P}$ is symmetric. Taking the derivative with respect to each x_k, we get

$$\begin{aligned}\frac{\partial f(\boldsymbol{x})}{\partial x_k} &= 2P_{kk}x_k + \underbrace{\boxed{2\sum_{k<j} P_{kj}x_j}}_{\substack{\text{derivative of}\\ i=k \text{ terms}}} + \underbrace{\boxed{2\sum_{i<k} P_{ik}x_i}}_{\substack{\text{derivative of}\\ j=k \text{ terms}}} + q_k \\ &= 2P_{kk}x_k + 2\sum_{k<j} P_{kj}x_j + 2\sum_{k>i} P_{ki}x_i + q_k \\ &= 2\sum_{i=1}^{n} P_{ki}x_i + q_k \\ &= 2\begin{bmatrix} P_{k1} & \cdots & P_{kn}\end{bmatrix}\boldsymbol{x} + q_k.\end{aligned}$$

Therefore, the gradient of $f(\boldsymbol{x})$ is given by

$$\nabla f(\boldsymbol{x}) = 2\mathbf{P}\boldsymbol{x} + \boldsymbol{q}.$$

Recall previous theorem: if $\boldsymbol{x}^*$ is a local minimum of $f(\boldsymbol{x})$, then $\nabla f(\boldsymbol{x}^*) = \mathbf{0}$. Thus,

$$\begin{aligned}\min \underbrace{\boxed{\|\mathbf{A}\boldsymbol{x} - \boldsymbol{y}\|}}_{\substack{\text{non-differentiable}\\ \text{at origin}}} &\Rightarrow \min \underbrace{\boxed{\|\mathbf{A}\boldsymbol{x} - \boldsymbol{y}\|^2}}_{f(\boldsymbol{x})} \\ &\Rightarrow \nabla f(\boldsymbol{x}^*) = \mathbf{0} \\ &\Rightarrow 2\underbrace{(\mathbf{A}^{\mathrm{T}}\mathbf{A})}_{\mathbf{P}}\boldsymbol{x}^* + \underbrace{(-2\mathbf{A}^{\mathrm{T}}\boldsymbol{y})}_{\boldsymbol{q}} = \mathbf{0} \\ &\Rightarrow \mathbf{A}^{\mathrm{T}}\mathbf{A}\boldsymbol{x}^* = \mathbf{A}^{\mathrm{T}}\boldsymbol{y}.\end{aligned}$$

- Previously, we obtained this formula via projection of $\boldsymbol{y}$ onto $\mathcal{R}(\mathbf{A})$.
- Later, we will learn that since $f(\boldsymbol{x})$ is convex, the equation $\nabla f(\boldsymbol{x}^*) = \mathbf{0}$ is both necessary and sufficient for global optimality (all local minima in this problem are global minima).
- Set of solutions to the least-squares problem: $\mathcal{S} = \{\boldsymbol{x}^* \mid \mathbf{A}^{\mathrm{T}}\mathbf{A}\boldsymbol{x}^* = \mathbf{A}^{\mathrm{T}}\boldsymbol{y}\}$.
- We can find the best solution with the minimum length or energy:

$$\begin{aligned}\min_{\boldsymbol{x}\in\mathbb{R}^n} \quad & \|\boldsymbol{x}\| \\ \text{subject to} \quad & \boldsymbol{x} \in \mathcal{S}.\end{aligned}$$

- To do so, we need notions of *eigenvalues* and *singular values*.

Given $\mathbf{A} \in \mathbb{R}^{n\times n}$, we denote its determinant as $\det(\mathbf{A})$. Example:

$$\det\left(\begin{bmatrix}2 & 4\\ 5 & 1\end{bmatrix}\right) = 2\times 1 - 4\times 5 = -18.$$

Consider a variable $\lambda \in \mathbb{R}$.

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- $P(\lambda) = \det(\lambda \mathbf{I}_n - \mathbf{A})$ is called the *characteristic polynomial* of $\mathbf{A}$. Here $\mathbf{I}_n$ is the $n \times n$ identity matrix.
- $P(\lambda)$ is of degree n, so it has n zeros:

$$P(\lambda) = (\lambda - \lambda_1)(\lambda - \lambda_2) \cdots (\lambda - \lambda_n).$$

  $\lambda_1, \ldots, \lambda_n$ are complex numbers showing zeros or roots of $P(\lambda)$. (Note: the coefficient of λ^n in $P(\lambda)$ is $+1$.)
- $\lambda_1, \ldots, \lambda_n$ are called the *eigenvalues* of $\mathbf{A}$.
- $\lambda \in \mathbb{C}$ (set of complex numbers) is an eigenvalue iff there is a non-zero vector $\boldsymbol{u} \in \mathbb{C}^n$ (n-dimensional complex vector) such that $(\lambda \mathbf{I}_n - \mathbf{A})\boldsymbol{u} = \mathbf{0}$, or $\mathbf{A}\boldsymbol{u} = \lambda \boldsymbol{u}$.
- Note: $\boldsymbol{u}$ is called an eigenvector corresponding to λ, and it is not unique: $\alpha \boldsymbol{u}$ also works for all $\alpha \in \mathbb{R}$ and $\alpha \neq 0$.

Example 6. Let $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, then

$$\det(\lambda \mathbf{I} - \mathbf{A}) = (\lambda - 1)(\lambda - 1) = 0 \quad \Rightarrow \quad \lambda_1 = \lambda_2 = 1 \text{ (repeated eigenvalues).}$$

Let $\boldsymbol{u}$ be an eigenvector corresponding to $\lambda = 1$, then

$$\mathbf{A}\boldsymbol{u} = \lambda \boldsymbol{u} \Rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \lambda \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$
$$\Rightarrow \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{ or } \begin{bmatrix} 0 \\ 1 \end{bmatrix} \text{ (or any linear combination of them).}$$

■

Example 7. Let $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, then

$$\det(\lambda \mathbf{I} - \mathbf{A}) = (\lambda - 1)(\lambda - 1) \quad \Rightarrow \quad \lambda_1 = \lambda_2 = 1.$$

Let $\boldsymbol{u}$ be an eigenvector corresponding to $\lambda = 1$, then

$$\mathbf{A}\boldsymbol{u} = \lambda \boldsymbol{u} \Rightarrow \begin{cases} u_1 + u_2 = \lambda u_1, \\ u_2 = \lambda u_2 \end{cases}$$
$$\Rightarrow u_2 = 0 \text{ and } u_1 \text{ is arbitrary}$$
$$\Rightarrow \text{eigenvectors: } \begin{bmatrix} u_1 \\ 0 \end{bmatrix}.$$

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- In Example 6, span of eigenvectors has dimension 2.

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- In Example 7, span of eigenvectors has dimension 1.

Consider $\mathbf{A} \in \mathbb{R}^{n \times n}$ with eigenvalues $\lambda_1, \ldots, \lambda_n$.

- Let $\boldsymbol{u}^{(1)}, \boldsymbol{u}^{(2)}, \ldots, \boldsymbol{u}^{(n)}$ be arbitrary eigenvectors, each associated with one eigenvalue.
- Assume $\boldsymbol{u}^{(1)}, \ldots, \boldsymbol{u}^{(n)}$ are linearly independent, i.e., $\text{span}(\boldsymbol{u}^{(1)}, \ldots, \boldsymbol{u}^{(n)})$ has dimension n.
- Then, $\mathbf{A}$ can be decomposed as $\mathbf{U}\mathbf{\Lambda}\mathbf{U}^{-1}$, where

$$\mathbf{U} = \begin{bmatrix} \boldsymbol{u}^{(1)} & \cdots & \boldsymbol{u}^{(n)} \end{bmatrix} \text{ and } \mathbf{\Lambda} = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix}.$$

- Since (by assumption) $\boldsymbol{u}^{(1)}, \ldots, \boldsymbol{u}^{(n)}$ are linearly independent, $\mathbf{U}$ is invertible.
- In this case, $\mathbf{A}$ is called a *diagonalizable* matrix.
- If $\lambda_1, \ldots, \lambda_n$ are all distinct (not repeated), $\mathbf{A}$ is always diagonalizable.
- If $\mathbf{A}$ has repeated eigenvalues, check Theorem 3.4 in the textbook that explains when such linearly independent eigenvectors $\boldsymbol{u}^{(1)}, \ldots, \boldsymbol{u}^{(n)}$ exist.

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